

Nicholas Niako, MS

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Education

Ph.D., Biostatistics (Student), University of South Carolina *2023–Present*

M.S., Mathematics, University of Texas Rio Grande Valley (UTRGV) *2023*

Thesis: *Effects of Missing Data Imputation Methods on Univariate Time Series Forecasting with ARIMA and LSTM*

Advisor: Kristina Vatcheva, Ph.D.

B.S., Statistics, Kwame Nkrumah University of Science and Technology (KNUST) *2018*

Research Experience

Graduate Research Assistant *2023– Present*

Department of Exercise Science, University of South Carolina

- Leading the development of an R package that integrates wearable sensor data with time-use diaries to streamline visualization and behavioral data analysis.
- Developed custom R functions to detect and clean duplicate GGIR output segments, improving data quality and analytical reliability.
- Standardized data workflows through clear documentation and collection templates that improved consistency between research projects.
- Contributed to manuscripts, reports, and presentations, translating analytical findings into accessible insights for scientific dissemination.

Research Assistant *2018–2020*

Department of Pathology, School of Medicine and Dentistry, KNUST

- Performed statistical analysis and managed Histopathological datasets to support clinical research studies.
- Prepared and cleaned research data to support accurate and efficient statistical analysis.
- Contributed to the preparation of research manuscripts and internal reports.
- Maintained accurate records and supported adherence to ethical guidelines and institutional protocols during data collection and handling.

Publications

1. **Niako, N.**, Melgarejo, J. D., Maestre, G. E., & Vatcheva, K. P. (2024). Effects of missing data imputation methods on univariate blood pressure time series data analysis and forecasting with ARIMA and LSTM. *BMC Medical Research Methodology*, 24(1), 320.

2. White III, J. W., Weaver, R. G., Finnegan, O., Cepni, A. B., Zhu, X., von Klinggraeff, L., Parker, H., Bastyr, M., **Niako, N.**, de Zambotti, M., Welk, G. J., Nelakuditi, S., Wang, Y., Burkart, S., & Ghosal, R. (2024). MACHINE LEARNING MODELS FOR PREDICTING LABORATORY-BASED PHYSICAL ACTIVITY TYPE FROM CONSUMER WEARABLES IN CHILDREN. *International Journal of Exercise Science: Conference Proceedings*, 16(3), Article 185.
3. Maxwell, A. K., Weaver, R. G., Armstrong, B., White, J., Finnegan, O., Cepni, A., Zhu, X., Von Klinggraeff, L., Parker, H., Bastyr, M., **Niako, N.**, Adams, E., Burkart, S., Beets, M., & Chandrashekhar, M. V. S. (2024). DETERMINING PULSE-RATE FROM WRIST PLACED ACCELEROMETRY IN CHILDREN IN ORDER TO IMPROVE ESTIMATES OF SLEEP. *International Journal of Exercise Science: Conference Proceedings*, 16(3), Article 81.
4. Weaver, R. G., White III, J., Finnegan, O., Cepni, A., Zhu, X., Von Klinggraeff, L., Parker, H., Bastyr, M., **Niako, N.**, Adams, E., Armstrong, B., Burkart, S., & Beets, M. (2024). OPENING THE BLACK BOX: FEASIBILITY OF COLLECTING RAW ACCELEROMETRY DATA FROM CONSUMER WEARABLE DEVICES FOR RESEARCH WITH CHILDREN. *International Journal of Exercise Science: Conference Proceedings*, 16(3), Article 4.
5. Ossei, P. P., **Niako, N.**, Ayibor, W. G., Asante, E., Safo, F. K., & Safowaa, A. (2022). Profile of suicide within the northern part of Ghana: A decade under review. *South African Journal of Psychiatry*, 28, 5.
6. Ossei, P. P. S., Agyeman-Duah, E., Ayibor, W. G., **Niako, N.**, & Safo, K. F. (2021). Latent Medical Conditions of Drivers Involved in Road Traffic Collisions in Ghana: Implication from Autopsy Findings. *Clinical Pathology*, 14, 2632010X211043871.
7. Ossei, P. P. S., Owusu, I. K., Owusu-Asubonteng, G., Ankobea-Kokroe, F., Ayibor, W. G., & **Niako, N.** (2020). Prevalence of Venous Thromboembolism in Kumasi: A Postmortem-Based Study in a Tertiary Hospital in Ghana. *Clinical Medicine Insights: Circulatory, Respiratory and Pulmonary Medicine*, 14, 1179548420956364.
8. Ossei, P. P. S., Agagli, B. M., Ayibor, W. G., **Niako, N.**, & Asante, E. (2020). Histological profile of kidney malignancies at a tertiary hospital in the Ashanti Region of Ghana; A 9-year review. *Journal of Cancer Research and Practice*, 7(2), 67.
9. Ossei, P. P. S., **Niako, N.**, Ayibor, W. G., Asante, E., Safo, K. F., Mensah, A. S., Owusu, E., & Appiah, K. L. (2020). Review of gunshot fatalities in the Northern part of Ghana; a 6-year forensic autopsy-based study. *Journal of Forensic and Legal Medicine*, 69, 101889.
10. Ossei, P. P. S., Appiah-Kubi, A., Ankobea-Kokroe, F., Owusu-Asubonteng, G., Ayibor, W. G., Aninkora, O. K., Taylor, J., Fenteng, E. A., Agyemang-Duah, E., Agagli, B. M., & **Niako, N.** (2020). The culture of herbal preparations among pregnant women: a remedy or a suicide potion? A Case Report and Mini Review. *Case Reports in Pathology*, 2020.
11. Ossei, P. P. S., Agagli, B. M., Ayibor, W. G., **Niako, N.**, & Asante, E. (2019). Trend analysis and economic effect of RTA deaths on dependency ratio in Ghana. *Journal of Economics and International Finance*, 11(7), 85–90.

Selected Presentations

Oral

1. An R package for Data Visualization of Time Diaries and Accelerometry. *International Society of Behavioral Nutrition and Physical Activity, Auckland, NZ Jun. 2025*
2. Performance evaluation of predicting children's sleep via long short-term memory (LSTM) models. *International Society of Behavioral Nutrition and Physical Activity, Auckland, NZ. Jun. 2025*
3. Opening the Black Box: Feasibility of collecting raw accelerometry data from consumer wearable devices for research with children. *Southeastern ACSM, Greenville, SC, Feb. 2024*
4. Effect of missing data imputation methods on univariate time series forecasting with ARIMA and LSTM. *SPIE Defense+ Commercial Sensing, Orlando, FL. May 2023*
5. Effect of missing data imputation methods on univariate time series forecasting with ARIMA and LSTM. *7th Coastal Bend Mathematics and Statistics Conference, Brownsville, TX. Apr. 2023*

Poster

1. Machine Learning Models for Predicting Free-living Activity Type from Raw Accelerometry and Heart Rate in Children *International Society of Behavioral Nutrition and Physical Activity, Cádiz, Spain May. 2026*
2. Machine Learning Models for Predicting Laboratory-Based Physical Activity Type from Consumer Wearables in Children. *Southeast Chapter of the American College of Sports Medicine (SEACSM), Greenville, SC. Feb. 2024*
3. Evaluation of missing data imputation techniques in univariate time series data forecasting with ARIMA. *ASA Conference on Statistical Practice, San Francisco, CA. Feb. 2023*
4. Firearm fatalities in the northern part of Ghana. *SIAM Texas-Louisiana Section, South Padre Island, TX. Nov. 2021*

Teaching Experience

Graduate Teaching Assistant

2025

Department of Epidemiology & Biostatistics, University of South Carolina

- Assisted instruction (probability, hypothesis testing, statistical inference)
- Presented guest lectures and facilitated discussions under faculty supervision.
- Collaborated on course materials and pedagogical refinements.

Graduate Teaching Assistant

2021–2022

Department of Mathematics, University of Texas Rio Grande Valley

- Tutored students in Calculus II; led interactive labs; facilitated academic discussions.
- Managed inclusive classroom dynamics.
- Graded for Real Analysis I, Introduction to Mathematical Software, and Calculus II with timely feedback.

Service

- **Graduate Student Experience Panelist**
StatFest, American Statistical Association (Virtual), 2025
- **Graduate School Survival Panelist**
Diversity Mentoring Program, Joint Statistical Meetings (JSM), Nashville, TN, 2025
- **Vice President**
American Mathematical Society Graduate Student Chapter, University of Texas Rio Grande Valley, 2022–2023
- **Undergraduate Course Mentor**
Kwame Nkrumah University of Science and Technology (KNUST), 2016–2017

Professional Memberships

- American Statistical Association (ASA)
- American Mathematical Society
- Association of African Quantitative Scientists
- Western North American Region of the International Biometric Society

Honors & Awards

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| 1. Graduate Student Travel Award | 2025 |
| 2. Travel Award, Diversity Mentoring Program, JSM | 2025 |
| 3. Arnold School of Public Health Doctoral Fellowship | 2023, 2024, 2025 |
| 4. Outstanding Graduate Thesis Award, UTRGV | 2023 |
| 5. Best Graduate Student Presentation, College of Science Annual Conference, UTRGV | 2023 |
| 6. Dean Graduate Assistantship Award, UTRGV | 2021, 2022, 2023 |

Skills

Proficient in R, Stata, LaTeX, Microsoft Office Suite;
Familiar with SAS, Python, MATLAB, SQL.

Technical Skills-

Proficient: R, Python, Stata, LaTeX, Linux/Unix, SLURM, Microsoft Office Suite

Working Knowledge: SAS, MATLAB, SQL

Tools: Git/GitHub, Jupyter Notebooks, High-Performance Computing (HPC)